

Munkres 4

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1 Introduction

In $\mathbb{R}^n$, let $\{x_n\}$ be a set of n points, $\{\epsilon_n\}$ be a set of n positive reals. Then

$$\bigcup_{i \in [n]} D(x_i, \epsilon_i) \subsetneq \bigcup_{i \in [n]} \overline{D}(x_i, \epsilon_i).$$

Let $(A_n), (B_n)$ be increasing chains of sets such that $A_n \cap B_n = \emptyset$ for all $n \in N$. Then is $\bigcup A_n \cap \bigcup B_n = \emptyset$? If otherwise, $A_n \cap B_m \neq \emptyset$ for some n, m . WLOG $n \leq m$, so $A_n \cap B_n = \emptyset$. N can be replaced by any totally ordered set.

If $f : X \rightarrow Y$ is closed continuous surjective and X is normal, then Y is normal.

If X, Y are metrizable, then so is $X \times Y$.

Every linear continuum L is normal. Proof sketch (ignoring edge cases):

(1) L is locally connected. Suppose $x \in (a, b)$. Pick $a < u < x$ and $x < v < b$, so $x \in (u, v) \subset [u, v] \subset (a, b)$, and $\overline{(u, v)} = [u, v]$.

(2) Consider $K = L - A \cup B \neq \emptyset$ (since L is connected). Since L is locally connected and K is open, we can partition K into a collection $\mathcal{A}$ of open connected components. Open connected components in a linear continuum are open intervals.

(3) Suppose $I = (u, v) \in \mathcal{A}$. Note $u \notin K$ as otherwise $[u, v]$ is connected and larger than I . Hence $u \in A$ or $u \in B$. Similarly, $v \in A$ or $v \in B$. If u, v are in different sets (wrt A, B), pick $x \in I$. Denote by S the set of all such x .

(4) Suppose $B \subset S$ is a nonempty and bounded below. Suppose B has no minimum. Let $b = \inf(B)$. Suppose for every $u > b$, $[b, u) \cap B \neq \emptyset$. In particular, there exist at least two x 's in this intersection, so $[b, u) \cap A \neq \emptyset$ and $[b, u) \cap B \neq \emptyset$. It follows $b \in A \cap B = \emptyset$, cont.

(5) For every $x \in S$, assuming that x is not the largest element of S , there

exists the successor x^+ of x . $I = (x, x^+)$ cannot intersect both A and B . Consider I as $x \in S$ and assign I to either a covering of A or a covering of B . If A, B were to have an unbounded connected component, extra care is needed.

Urysohn's Lemma. Let X be a normal space, A, B disjoint nonempty closed subsets. Let $U_1 = X - B$ be open. Since $A \subset X - B$, by normality, there exists $\overline{U_0(A)} \subset U_1$. Index the rationals in $(0, 1)$ by N . Suppose at step $n \geq 1$ we have the chain of $n + 1$ open sets indexed by a finite subset of $\mathbb{Q}$:

$$U_0 \subset U_{q_1} \subset U_{q_2} \subset \cdots \subset U_{q_n} \subset U_1$$

with the property that if $p < q$ occur as indices then $\overline{U_p} \subset U_q$. Pick p, q to be extremal indices such that $p < p_n < q$ and pick using normality

$$\overline{U_p} \subset U_{p_n} \subset \overline{U_{p_n}} \subset U_q.$$

By principle of inductive definition, there exists a chain of open sets U_p indexed by $\mathbb{Q}_{[0,1]}$ such that if $0 \leq p < q \leq 1$, then $\overline{U_p} \subset U_q$ and also $A \subset U_0, U_1 = X - B$.

Define the function $f : X \rightarrow [0, 1]$ by the following rule:

- (1) If $x \in B$, let $f(x) = 1$.
- (2) If $x \notin B$, let $S \subset \mathbb{Q}_{[0,1]}$ denote the set of rationals q such that $x \in U_q$. $S \neq \emptyset$ since $x \in U_1$, so let us define $f(x) = \inf S$.

I claim f is continuous. Let us proceed with some lemmas.

- (1) If $f(x) < q$ for $q \in \mathbb{Q}_{[0,1]}$, then $x \in U_q$. If $x \notin U_q$, then by construction (of the chain), $x \notin U_p$ for all $p < q$, so $f(x) \geq q$, cont.
- (2) If $f(x) > q$ for $q \in \mathbb{Q}_{[0,1]}$, then $x \notin \overline{U_q}$. Let $y = f(x)$. Pick $q < r < y$ for $r \in \mathbb{Q}_{[0,1]}$. If $x \in U_r$, then $y = f(x) \leq r$, cont. Since $U_r \supset \overline{U_q}$, $x \notin \overline{U_q}$.

Now suppose $f(x) = y \in (0, 1)$. Pick $\epsilon > 0$ sufficiently small st $(y - \epsilon, y + \epsilon) \subset (0, 1)$. Pick $p, q \in \mathbb{Q}_{[0,1]}$ such that $y - \epsilon < p < y < q < y + \epsilon$. By lemmas 1, 2, $x \in U(x) = U_q - \overline{U_p}$. $U(x)$ is open and if $x' \in U(x)$, since $U(x) \subset U_q$, $f(x') \leq q < y + \epsilon$. Next, if $f(x') \leq y - \epsilon$, then $f(x') < p$, so $x' \in U_p$, cont. It follows $f(U(x)) \subset (y - \epsilon, y + \epsilon)$.

Suppose $y = 0$. Pick $1 > \epsilon > q > 0$ for $q \in \mathbb{Q}_{[0,1]}$. If $x \in U_q$, then $y \leq q < \epsilon$, so $f(U_q) \subset [0, \epsilon)$. $x \in U_q$ since $y = 0 < q$.

Suppose $y = 1$. Pick $1 > \epsilon > 0$ and $1 - \epsilon < q < y = 1$. Since $y > q$, $x \notin \overline{U_q}$, so $x \in X - \overline{U_q}$. Let $x' \notin \overline{U_q}$. If $f(x') \leq 1 - \epsilon$, then $x' \in U_q \subset \overline{U_q}$, cont.

Hence, f is continuous.

Suppose $A_1, A_2, \dots, A_n$ are nonempty sets such that for $i \neq j$, A_i, A_j are separated by a continuous function $f_{(i,j)} : X \rightarrow [0, 1]$ such that $f(A_i) = \{0\}$ and $f(A_j) = \{1\}$. Then there exists a continuous function $f : X \rightarrow \mathbb{R}$ such that $f(A_i) = i$ for all $i \in [n]$.

Proof. Define $g_i : X \rightarrow \mathbb{R}$ by $g_i = \prod_{j \neq i} f_{(j,i)}$. Then g_i is continuous and $g_i(A_j) = \{0\}$ for $j \neq i$ and $g_i(A_i) = \{1\}$. Define $f = \sum_{i \in [n]} i g_i$. Then f is continuous and $f(A_i) = i$.

If X, Y are $T_{3\frac{1}{2}}$, then so is $X \times Y$.

(1) X, Y are T_1 iff $X \times Y$ is T_1 . In the forward direction, suppose $(x_0, y_0) \in X \times Y$ is fixed and $(x, y) \neq (x_0, y_0)$ is arbitrary. Then at least one of x, y doesn't equal the corresponding x_0, y_0 . WLOG $x \neq x_0$. Since X is T_1 , pick $U(x)$ disjoint from x_0 . Then $U \times Y$ is disjoint from (x_0, y_0) .

Now suppose $X \times Y$ is T_1 . Fix x_0, y_0 in X, Y , resp. Pick $x \neq x_0$ to be arbitrary. There exists $U(x) \times V(y)$ disjoint from (x_0, y_0) , so $x_0 \notin U(x)$ or $y_0 \notin V(y)$. The latter is impossible, so X is T_1 . Similarly for Y .

X, Y are $T_{3\frac{1}{2}}$, so they're also T_1 , and so is $X \times Y$ by Lemma 1. Suppose $C \subset X \times Y$ is closed. If $|X| = 1$ or $|Y| = 1$, then (WLOG) $X \times Y \cong X$, so $X \times Y$ would be $T_{3\frac{1}{2}}$. Else, $|X|, |Y| \geq 2$. If $U(x) = X$, take $U'(x) = U(x) - \{x'\}$ for some $x' \neq x$, else take $U'(x) = U(x)$. Now $X - U'(x) \neq \emptyset$ is closed. By $T_{3\frac{1}{2}}$ of X , pick continuous $f : X \rightarrow I$ st $f(x) = 1$ and $f(X - U'(x)) = \{0\}$. Pick g similarly for Y . Now take $h = fg : X \rightarrow I$ given by $h(x, y) = f(x)g(y)$, which is continuous being the composition $(x, y) \rightarrow (f(x), g(y)) \rightarrow f(x)g(y)$. Further, $h(x, y) = f(x)g(y) = 1$ and

$$f((X - U'(x)) \times Y \cup X \times (Y - V'(y))) = f(X \times Y - U'(x) \times V'(y)) = 0,$$

and the set $X \times Y - U'(x) \times V'(y)$ contains C , AWD.

If $\{X_\alpha\}, \{Y_\alpha\}$ are collections of spaces st $f_\alpha : X_\alpha \rightarrow Y_\alpha$ is continuous, is $f = \prod f_\alpha : \prod X_\alpha \rightarrow \prod Y_\alpha$ cont? Consider basis element $U = \prod U_\alpha$ of $\prod Y_\alpha$ where all but finitely many U_α are X_α . Then $f^{-1}(U) = \prod f_\alpha(U_\alpha)$, which is open.

Suppose $x \notin C$ for $C \subset \prod X_\alpha$ closed, each X_α being $T_{3\frac{1}{2}}$ and having at least two elements. Pick $U(x)$ open st $U = \prod U_\alpha$, all but finitely many are X_α . If $U_\alpha = X_\alpha$, take $f_\alpha : X \rightarrow I$ to be $f(X_\alpha) = \{1\}$. Pick $U'(x_\alpha)$ as before st $D_\alpha = X_\alpha - U'(x_\alpha) \neq \emptyset$. By $T_{3\frac{1}{2}}$, pick $f_\alpha : X_\alpha \rightarrow I$ st $f_\alpha(x_\alpha) = 1$ and $f_\alpha(D_\alpha) = \{0\}$.

Now take $f : \prod X_\alpha \rightarrow \prod Y_\alpha$ to be $f(x) = \prod f_\alpha(x_\alpha)$, well-defined since all

but finitely many α have $f_\alpha \equiv 1$, continuous by previous proposition. $f(x) = 1$. Let the set of α for $f_\alpha \not\equiv 1$ be K (K finite) ... definition nonsense ... same construction for two spaces should work.

Let X be regular second-countable. Then X is perfectly normal. It was shown X is normal.

Let $\mathcal{B}$ be a countable basis and $A \subset X$ be closed. Let $\mathcal{B}' = \{B \in \mathcal{B} : \overline{B} \cap A = \emptyset\}$. I claim $\bigcup \mathcal{B}' = X - A$. If $x \in \mathcal{B}'$, then $x \in B$ for some $\overline{B} \cap A = \emptyset$, so $x \notin A$. Now suppose $x \notin A$. Since A is closed, pick $U(x) \cap A = \emptyset$. By regularity, pick $\overline{V(x)} \subset U(x)$. Pick basis $B(x) \subset V(x)$. Then $\overline{B} \subset \overline{V} \subset U$, so $\overline{B} \cap A = \emptyset$. Hence, $x \in \bigcup \mathcal{B}'$, showing the result.

Since $\mathcal{B}' \subset \mathcal{B}$ is countable (or finite, but this gives a similar case), index $\mathcal{B}'$ by N . Define $A_n = X - \bigcup_{k=1}^n \overline{B'_k}$, so $\{A_n\}$ is a descending chain of open sets. Since

$$X - A = \bigcup_{n \in N} B'_n \subset \bigcup_{n \in N} \overline{B'_n} \subset X - A,$$

we have

$$\bigcap_{n \in N} A_n = X - \bigcup_{n \in N} \overline{B'_n} = A.$$

Hence, A is G_δ and X is perfectly normal.

Does there exist a continuous map $\mathbb{R} \rightarrow \mathbb{R}^2$?

There exists a countable collection of continuous functions $\{f_n\}$ such that for all $x \in X$ and $U(x)$, there exists $f_n : X \rightarrow I$ for which $f(x) > 0$ and $f(X - U) = \{0\}$.

Proof: Let $\mathcal{B}$ be a countable basis for X . For all $B_n \in \mathcal{B}$, by perfect normality construct continuous $f_n : X \rightarrow I$ such that f_n vanishes exactly on $X - B$. I claim $\{f_n\}$ is the desired collection.

If $x \in X$ and $U(x)$ is open, pick $B_n(x) \subset U(x)$. Then f_n has $f(x) > 0$ and $f(X - U) \subset f(X - B) = \{0\}$, as desired.

Munkres proof: Let $\mathcal{B}$ be a countable basis for X indexed by N . For all $(m, n) \in N^2$, whenever $\overline{B_m} \subset B_n$, by Urysohn lemma construct $f_{m,n} : X \rightarrow I$ such that $f(\overline{B_m}) = \{1\}$ and $f(X - B_n) = \{0\}$. The set of possible $f_{m,n}$ is countable since N^2 is countable. Suppose $x \in X$ and $U(x)$ is open. Pick $B_n(x) \subset U$. By regularity of X , pick $\overline{B_m(x)} \subset B_n$. Then $f_{m,n}$ is the desired function since $f_{m,n}(x) = 1$ and $f_{m,n}(X - U) \subset f_{m,n}(X - B_n) = \{0\}$.

Let $\{f_n\}$ be a collection of continuous functions with the above property. Define $F : X \rightarrow \mathbb{R}^\omega$ by $F(x) = (f_1(x), f_2(x), f_3(x), \dots)$. Then F is an embedding of

X into $\mathbb{R}^\omega$.

Proof: Since $\mathbb{R}^\omega$ has the product topology and each f_n is continuous, F is continuous. If $x \neq y$, pick $U(x)$ disjoint to y . Then for some $n \in N$, $f_n(x) > 0$ and $f_n(y) = 0$, so $F(x) \neq F(y)$. That is, F is injective.

I claim to show that F restricted to $F(X)$ is an open map, it suffices to show that if $y \in F(U)$ with U open, there exists $V(y)$ open in $\mathbb{R}^\omega$ such that $F^{-1}(V) \subset U$. If this is the case, $V \cap F(X) = F(F^{-1}(V)) \subset F(U)$, so $F(U)$ would be open in $F(X) \cap \mathbb{R}^\omega$.

Let $y \in F(U)$, so $y = F(x)$ for some $x \in U$. There exists $n_0 \in N$ such that $f_{n_0}(x) > 0$ and $f_{n_0}(X - U) = \{0\}$. Let $V_n = \mathbb{R}$ for $n \neq n_0$ and $V_{n_0} = \mathbb{R}^+$. Let $V = \prod_{n \in N} V_n$. Then $F^{-1}(V) = f_{n_0}^{-1}(\mathbb{R}^+) \subset U$.

Book proof 2: Take $f_n \leq 1/n$ for all $n \in N$. Define F the same way. Then we claim $F : X \rightarrow \mathbb{R}^\omega$ in the uniform topology is continuous. Let $x \in X$ and $y = F(x)$. Pick $T = \prod_{n=1}^\infty (y_n - \epsilon, y_n + \epsilon)$ to be a tube containing y . There exists $n_0 \in N$ such that $\frac{1}{n_0} < \epsilon$. Suppose $y' \in T$. Then for all $n \geq n_0$, $|y_n - y'_n| \leq \frac{1}{n} < \epsilon$, T covers $f(X)$ for $n \geq n_0$. Hence we have

$$f^{-1}(T) = \bigcap_{n < n_0} f_n^{-1}((y_n - \epsilon, y_n + \epsilon)),$$

so $f^{-1}(T)$ is open in X , being a finite intersection of open sets. Any open set in $\mathbb{R}^\omega$ is a union of these tubes, so f is continuous.

The $\{f_n\}$ retain the property that for all $x \in X$, $U(x)$ open, there exists $n \in N$ for which $f_n(x) > 0$ and $f(X - U) = \{0\}$. Hence, F is injective by the same argument.

By the same argument, F restricted to $F(X)$ is a homeomorphism.

Show that a locally metrizable compact Hausdorff space X is metrizable.

Since X is compact Hausdorff, X is regular. Let $x \in X$ and pick $U(x)$ to be metrizable. By regularity, pick $\overline{V(x)} \subset U$. Since X is compact, $\overline{V}$ is compact. Since U is metrizable, $\overline{V}$ is metrizable. Since $\overline{V}$ is compact metrizable, $\overline{V}$ is second countable, so V is also second countable. By compactness of X , pick a finite subcovering $\mathcal{A}$ of $\{V(x)\}_{x \in X}$. If $Y, Z \subset X$ are open subspaces, and $\mathcal{B}_X, \mathcal{B}_Y, \mathcal{B}_Z$ are bases for Y, Z , then $\mathcal{B}_X \cup \mathcal{B}_Y$ is a basis for $Y \cup Z$ as a subspace of X . If $V_1, V_2 \in \mathcal{A}$, then V_1, V_2 have countable bases, and so does $V_1 \cup V_2$. By an inductive argument, $X = \bigcup \mathcal{A}$ has a countable basis, so X is regular second countable, and therefore metrizable.

Show that a regular locally metrizable Lindelof space X is metrizable.

Let $x \in X$ and $U(x)$ be metrizable. Pick $\overline{V(x)} \subset U$, so $\overline{V}$ is metrizable. Since X is Lindelof and $\{V(x)\}_{x \in X}$ covers X , pick a countable subcover $\mathcal{A}$.

If Y is metrizable Lindelof, then Y is second countable. Proof: Let $n \in \mathbb{N}$. The collection $\{U(y, 1/n)\}_{y \in Y}$ covers Y , so pick a countable subcovering $\mathcal{B}_n$. I claim that $\mathcal{B} = \bigcup_{n \in \mathbb{N}} \mathcal{B}_n$ is a countable basis for Y . Let $y_0 \in Y$ and consider $U(y_0, \epsilon)$. Pick $2/n < \epsilon$. $\mathcal{A}_n$ covers Y , so there exists some open $D \in \mathcal{A}_n$ (all of which have diameter at most $2/n$) containing y_0 . Then $D \subset U(y_0, \epsilon)$, so $\mathcal{B}$ is indeed a countable basis for Y .

Now suppose $x_0 \in X$. There exists some $V \in \mathcal{A}$ such that $x_0 \in V$. Since $\overline{V}$ is a closed subset of Lindelof space X , $\overline{V}$ is Lindelof. By the claim, $\overline{V}$ is second countable, so V is second countable as well. X is a countable union of second countable open subspaces, and is therefore also second countable. A regular second countable space is metrizable.

If X is second countable locally compact Hausdorff, then the one point compactification Y of X is metrizable.

Let $\mathcal{B}$ be a countable basis for X . For all $x \in X$, pick $x \in B$ for some B . By local compactness, pick $B'(x) \subset C(x) \subset B$, where $B' \in \mathcal{B}$ and C is compact. Denote by $\mathcal{C}$ a countable set of such C each containing a distinct B' (index over B' and pick exactly one $C \supset B'$). Then I claim that $\mathcal{B} \cup (Y - \mathcal{C})$ is a countable basis for Y .

The open sets of Y are either an open set of X or a set of the form $Y - D$, where D is compact. The first case is handled by the basis $\mathcal{B}$. For the second, for each $d \in D$, there exists $B'(d) \subset C(d)$, where $C \in \mathcal{C}$. Then the set $\{B'(d)\}_{d \in D}$ is an open covering of D . By compactness, pick a finite subcovering indexed by D' . Then $D \subset \bigcup_{d' \in D'} C(d')$, so $D = D \cap \bigcup_{d' \in D'} C(d')$.

Now

$$Y - D = (Y - D) \cup (Y - \bigcup_{d' \in D'} C(d')) = (Y - D) \cup \bigcap_{d' \in D'} (Y - C(d')).$$

$Y - D$ could be replaced by $X - D$ since ∞ is already counted in the intersection over D' . Since D is compact and X is Hausdorff, D is closed, so $X - D$ is open. Hence, $Y - D$ is generated by $\mathcal{B} \cup (Y - \mathcal{C})$. It follows that Y is second countable.

Y is second countable compact Hausdorff, and therefore metrizable. Conversely, if Y is compact metrizable, then Y is second countable. X is a subspace of Y and therefore also second countable.

Let $A \subset X$ be closed. Then the connected components of A are closed.

Let $C \subset A$ be a connected component. Let E range over the connected components of A not equal to C . Then

$$\overline{C} \cap (A - C) = \overline{C} \cap \bigcup E = \bigcup \overline{C} \cap E = \emptyset,$$

since C is separable from each E . Then

$$\overline{C} \cap A = (\overline{C} \cap C) \cup (\overline{C} \cap (A - C)) = C,$$

so C is closed in A . Since A is closed in X , C is closed in X .

Shorter proof: Simply consider the connected components of X , let C be one of them. If C is connected, so is $\overline{C}$. Then $C \subset \overline{C} \subset C$, so $C = \overline{C}$, and C is closed.

Tietze Extension theorem. Let X be normal and $A \subset X$ be a closed subspace. Suppose $f : A \rightarrow [a, b]$ is continuous. Then there exists continuous $g : X \rightarrow [a, b]$ such that $g|_A = f$.

Munkres Proof: Take $f : X \rightarrow [-r, r]$ to be continuous. I claim there exists $h : X \rightarrow [-r/3, r/3]$ such that $|h(a) - f(a)| \leq 2r/3$ for all $a \in A$. Let $A_+ = \{a \in A : f(a) \geq r/3\}$ and $A_- = \{a \in A : f(a) \leq -r/3\}$ be disjoint closed sets. Construct $h : X \rightarrow [-r/3, r/3]$ such that $h(A_+) = \{r/3\}$ and $h(A_-) = \{-r/3\}$. If $a \in A_+ \cup A_-$, then we're done. Else, $-r/3 < f(a) < r/3$ and $|f(a) - h(a)| \leq |f(a)| + |h(a)| < r/3 + r/3 = 2r/3$.

Let $f_0 = 0$. I claim there exist functions $f_n : X \rightarrow \mathbb{R}$ such that

$$|f - f_0 - f_1 - \cdots - f_n| \leq (2/3)^n r$$

on A and $|f_n| \leq (2/3)^{n-1} \cdot r/3$ on X . We proceed by induction. The base case $n = 0$ holds. From the previous claim, pick $f_{n+1} : X \rightarrow \mathbb{R}$ such that $|f_{n+1}| \leq (2/3)^n \cdot r/3$ and

$$|f - f_0 - f_1 - \cdots - f_{n+1}| \leq (2/3)^n \cdot (2/3) = (2/3)^{n+1}.$$

Then the result follows.

By Weierstrass M-test, $g = \sum_{n \in \mathbb{N}} f_n$ converges uniformly to a continuous function $X \rightarrow \mathbb{R}$ since $\sum_{n \in \mathbb{N}} (2/3)^{n-1} \cdot (r/3)$ converges. By construction, g converges uniformly to f over A since $(2/3)^n \rightarrow 0$. It remains to show $g(X) \subset [-r, r]$. This is from calculating $\sum_{n \in \mathbb{N}} (2/3)^{n-1} \cdot r/3 = r$.

Let D_n be the open unit disc centered at $\vec{0}$ in $\mathbb{R}^n$. Then D_n is homeomorphic to $\mathbb{R}^n$ with the map $\vec{x} \mapsto \frac{\vec{x}}{1 - \|\vec{x}\|}$.

Suppose X is an m -manifold. To show m is unique: let $\mathcal{P}_m = \{\vec{x} \in \mathbb{R}^m : x_1 = 0\}$ and U be an open set in $\mathbb{R}^n$ with $n > m$. Let $A \sim \mathcal{P}_m$. Then $U - A$ is connected.

Is every connected 1-manifold homeomorphic to $\mathbb{R}$ or S_1 ?

Let $\{U_i\}$ be a finite open covering of the normal space X . Then there exists a partition of unity dominated by $\{U_i\}$.

Proof if X is compact: Claim: there exist closed $A_i \subset U_i$ such that $\{A_i\}$ covers X . For each $x \in X$, there exists U_i such that $x \in U_i$. By regularity, pick $\bar{V}(x) \subset U_i$. By compactness, pick a finite covering $\{V(x)\}$ of X . For each V pick a U_i such that $\bar{V} \subset U_i$. Fix the U_i and take the union of the $\bar{V}$ to be A_i . Then the $\{A_i\}$ give the desired construction.

Now we have $X - U_i \subset X - A_i$. By normality, pick $\overline{W(X - U_i)} \subset X - A_i$. By Urysohn Lemma, since $\bar{W}, A$ are disjoint, pick continuous $\phi_i : X \rightarrow I$ such that $\phi_i(\bar{W}) = \{0\}$ and $\phi_i(A) = \{1\}$. Then $\text{support}(\phi_i) \subset X - W = X - (X - U_i) = U_i$. Now take continuous $\phi : X \rightarrow \mathbb{R}$ such that $\phi = \phi_1 + \phi_2 + \dots + \phi_n$. Since the $\{A_i\}$ cover X , $\phi \geq 1$, so $\phi'_i = \phi_i/\phi : X \rightarrow I$ is continuous, $\sum \phi'_i = 1$, and $\text{support}(\phi'_i) = \text{support}(\phi_i) \subset U_i$, as desired.

Munkres proof (without compactness, for the first part): Suppose $\{U_i\}$ covers X . Then $A = X - (U_2 \cup U_3 \cup \dots \cup U_n) \subset U_1$. By normality, pick $\bar{V}_1(A) \subset U_1$. Then $V_1, U_2, U_3, \dots, U_n$ cover X . Repeat the process so that $V_1, V_2, \dots, V_n$ cover X , $V_i \subset \bar{V}_i \subset U_i$, so the $\bar{V}_i$ is a closed covering of X .

Let X be an m -manifold. Then X can be embedded in $\mathbb{R}^N$ Euclidean space for some N .

(1) X is normal. Let $x \in X$, so $U(x) \xleftrightarrow{h} V(y)$ for some $V(y) \subset \mathbb{R}^m$. Pick a basis element $V(y) = \prod (y_i - \epsilon_i, y_i + \epsilon_i) \subset U$. Then

$$C(y) = \prod [y_i - \epsilon_i/2, y_i + \epsilon_i/2] \supset W(y) = \prod (y_i - \epsilon_i/2, y_i + \epsilon_i/2)$$

are both contained in U with C compact, W open. Then $h^{-1}(W) \subset h^{-1}(C)$ are open, compact in X and both contain x .

Hence, X is locally compact. Since X is also Hausdorff, X is completely regular. Since X is regular and second countable, X is metrizable (in fact embedding in $\mathbb{R}^\omega$), though we only need that X is normal.

(2) There exists a finite partition of unity dominated by open $\{U_i\}$ covering X such that $U_i \xleftrightarrow{h_i} (0, 1)^m$. From (1), it is easy to see that for each $x \in X$, there exists $U(x) \xleftrightarrow{h} (0, 1)^m$ since $(0, 1)^m \xleftrightarrow{h} \prod (y_i - \epsilon_i, y_i + \epsilon_i)$. Since X is compact, there exists a finite covering $\{U_i\}$ indexed by $[n]$. Since X is normal, there exists a partition of unity dominated by the $\{U_i\}$.

(3) X can be imbedded in $\mathbb{R}^N$ for some N . Define $f_i : X \rightarrow \mathbb{R}^m$ by $f_i(x) =$

$\phi_i(x)h_i(x)$ if $x \in U_i$ and $f_i(x) = \vec{0}$ if $x \in \phi_i^{-1}(\{0\})$. f_i is well-defined since $\text{support}(\phi_i) \subset U_i$ and by gluing lemma. Let $F : X \rightarrow \mathbb{R}^N$ be the function

$$F(x) = (\phi_1(x), \phi_2(x), \dots, \phi_n(x), f_1(x), f_2(x), \dots, f_n(x)).$$

(3.1) F is injective. Suppose $F(x) = F(y)$. Since $\sum \phi_i = 1$, $\phi_i > 0$ for some i . Then $\phi_i(x) = \phi_i(y)$ and $\phi_i(x)h_i(x) = \phi_i(y)h_i(y)$. Since h_i is injective, $x = y$.

(3.2) If $U(x) \subset \mathbb{R}_{\geq 0}^m$ is open, there exists $\gamma > 0$ and $V(x)$ for which $(1 - \gamma, 1 + \gamma)V \subset U$, where

$$(1 - \gamma, 1 + \gamma)V = \{(k_1x_1, k_2x_2, \dots, k_mx_m) : x \in V, k_i \in (1 - \gamma, 1 + \gamma)\}.$$

Let $U = \prod (x_i - \epsilon, x_i + \epsilon)$ and $V = \prod (x_i - \delta, x_i + \delta)$. Pick any $\delta > 0$ smaller than ϵ . There exists $\gamma > 0$ for which

$$(1 - \gamma, 1 + \gamma) \subset \left(\frac{x_i - \epsilon}{x_i - \delta}, \frac{x_i + \epsilon}{x_i + \delta} \right)$$

for all $i \in [m]$. This is achievable since there are finitely many i and $\frac{x_i - \epsilon}{x_i - \delta} < 1 < \frac{x_i + \epsilon}{x_i + \delta}$. Then $(1 - \gamma, 1 + \gamma)V \subset U$.

(3.3) $F : X \rightarrow \mathbb{R}^N$ is an embedding. Fix $U(x) \subset X$ to be open. It suffices to show if $y = F(x)$ there exists $W(y) \subset \mathbb{R}^N$ open such that $F^{-1}(W) \subset U$.

First, pick i such that $\phi_i(x) > 0$, so $x \in \text{support}(\phi_i) \subset U_i$. Let $N(y) = h_i(U \cap U_i) \subset \mathbb{R}^m$ be open. Pick $\gamma > 0$ and $M(y)$ such that $(1 - \gamma, 1 + \gamma)M(y) \subset N$. Pick $\epsilon > 0$ such that

$$\left(\frac{\phi_i(x) - \epsilon}{\phi_i(x) + \epsilon}, \frac{\phi_i(x) + \epsilon}{\phi_i(x) - \epsilon} \right) \subset (1 - \gamma, 1 + \gamma).$$

Take W to be such that $F(y) \in W$ is equivalent to $\phi_i(y) \in (\phi_i(x) - \epsilon, \phi_i(x) + \epsilon)$ and $f_i(y) \in (\phi_i(x) - \epsilon, \phi_i(x) + \epsilon)M$ (possible by picking other open sets in the product to be free). Then

$$h_i(y) = f_i(y)/\phi_i(y) \in \left(\frac{\phi_i(x) - \epsilon}{\phi_i(x) + \epsilon}, \frac{\phi_i(x) + \epsilon}{\phi_i(x) - \epsilon} \right) M \subset (1 - \gamma, 1 + \gamma)M \subset N.$$

Hence, $y \in U \cap U_i$ (as h_i is injective), so F is indeed an embedding.